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Section: BCS-243D

Syllabus: Database Systems Lab

Syllabus Code: CL2005

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Q1)

Query:

```
SELECT e.first_name, d.department_name  
FROM HR.employees e  
INNER JOIN HR.departments d ON e.department_id = d.department_id;
```

Snapshot:

Worksheet

Query Builder

```
SELECT e.first_name, d.department_name
FROM HR.employees e
INNER JOIN HR.departments d ON e.department_id = d.department_id;
```

 Query Result x

    SQL | Fetched 50 rows in 0.138 seconds

	FIRST_NAME	DEPARTMENT_NAME
1	Jennifer	Administration
2	Pat	Marketing
3	Michael	Marketing
4	Sigal	Purchasing
5	Karen	Purchasing
6	Shelli	Purchasing
7	Den	Purchasing
8	Alexander	Purchasing
9	Guy	Purchasing
10	Susan	Human Resources
11	Kevin	Shipping
12	Jean	Shipping
13	Adam	Shipping
14	Timothy	Shipping
15	Ki	Shipping
16	Girard	Shipping
17	Douglas	Shipping
18	Vance	Shipping
19	Payam	Shipping
20	Renske	Shipping
21	Britney	Shipping
22	Jennifer	Shipping
23	Julia	Shipping
24	Curtis	Shipping

Q2)

Query:

```
SELECT e.first_name, e.last_name, d.department_name
```

```
FROM HR.employees e
```

```
LEFT OUTER JOIN HR.departments d ON e.department_id = d.department_id;
```

Snapshot:

Worksheet

Query Builder

```
SELECT e.first_name, e.last_name, d.department_name
FROM HR.employees e
LEFT OUTER JOIN HR.departments d ON e.department_id = d.department_id;
```

Query Result x

SQL | Fetched 50 rows in 0.01 seconds

	FIRST_NAME	LAST_NAME	DEPARTMENT_NAME
1	Jennifer	Whalen	Administration
2	Pat	Fay	Marketing
3	Michael	Hartstein	Marketing
4	Karen	Colmenares	Purchasing
5	Guy	Himuro	Purchasing
6	Sigal	Tobias	Purchasing
7	Shelli	Baida	Purchasing
8	Alexander	Khoo	Purchasing
9	Den	Raphaely	Purchasing
10	Susan	Mavris	Human Resources
11	Douglas	Grant	Shipping
12	Donald	OConnell	Shipping
13	Kevin	Feeney	Shipping
14	Alana	Walsh	Shipping
15	Vance	Jones	Shipping
16	Samuel	McCain	Shipping
17	Britney	Everett	Shipping
18	Sarah	Bell	Shipping
19	Randall	Perkins	Shipping
20	Timothy	Gates	Shipping
21	Jennifer	Dilly	Shipping
22	Kelly	Chung	Shipping
23	Anthony	Cabrio	Shipping
24	Julia	Dellinger	Shipping

Q3)

Query:

```
SELECT d.department_name, e.first_name, e.last_name
FROM HR.departments d
LEFT OUTER JOIN HR.employees e ON d.department_id = e.department_id;
```

Snapshot:

Worksheet		Query Builder	
		<pre>SELECT d.department_name, e.first_name, e.last_name FROM HR.departments d LEFT OUTER JOIN HR.employees e ON d.department_id = e.department_id;</pre>	
		Query Result x	
		SQL Fetched 50 rows in 0.007 seconds	
	DEPARTMENT_NAME	FIRST_NAME	LAST_NAME
1	Administration	Jennifer	Whalen
2	Marketing	Pat	Fay
3	Marketing	Michael	Hartstein
4	Purchasing	Sigal	Tobias
5	Purchasing	Karen	Colmenares
6	Purchasing	Shelli	Baida
7	Purchasing	Den	Raphaely
8	Purchasing	Alexander	Khoo
9	Purchasing	Guy	Himuro
10	Human Resources	Susan	Mavris
11	Shipping	Kevin	Feeney
12	Shipping	Jean	Fleaur
13	Shipping	Adam	Fripp
14	Shipping	Timothy	Gates
15	Shipping	Ki	Gee
16	Shipping	Girard	Geoni
17	Shipping	Douglas	Grant
18	Shipping	Vance	Jones
19	Shipping	Payam	Kaufling
20	Shipping	Renske	Ladwig
21	Shipping	Britney	Everett
22	Shipping	Jennifer	Dilly
23	Shipping	Julia	Dellinger
24	Shipping	Curtis	Davies

Q4)

Query:

```
SELECT e.first_name, j.job_title
FROM HR.employees e
JOIN HR.jobs j ON e.job_id = j.job_id;
```

Snapshot:

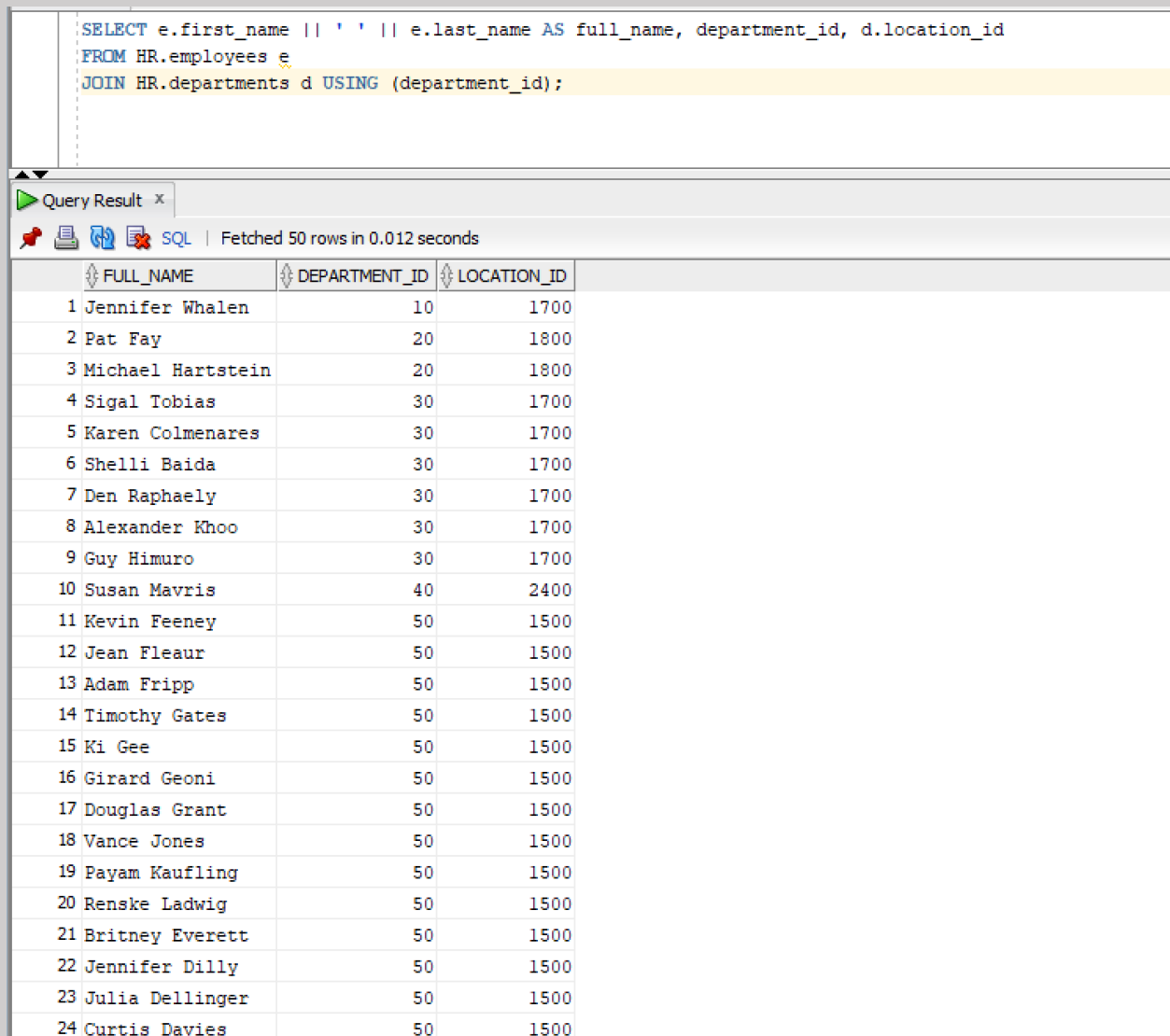
Worksheet		Query Builder
		<pre>SELECT e.first_name, j.job_title FROM HR.employees e JOIN HR.jobs j ON e.job_id = j.job_id;</pre>
Query Result x		
SQL Fetched 50 rows in 0.017 seconds		
	FIRST_NAME	JOB_TITLE
1	William	Public Accountant
2	Shelley	Accounting Manager
3	Jennifer	Administration Assistant
4	Steven	President
5	Neena	Administration Vice President
6	Lex	Administration Vice President
7	Jose Manuel	Accountant
8	Ismael	Accountant
9	Luis	Accountant
10	John	Accountant
11	Daniel	Accountant
12	Nancy	Finance Manager
13	Susan	Human Resources Representative
14	Valli	Programmer
15	Alexander	Programmer
16	David	Programmer
17	Bruce	Programmer
18	Diana	Programmer
19	Michael	Marketing Manager
20	Pat	Marketing Representative
21	Hermann	Public Relations Representative
22	Guy	Purchasing Clerk
23	Karen	Purchasing Clerk
24	Sigal	Purchasing Clerk

Q5)

Query:

```
SELECT e.first_name || ' ' || e.last_name AS full_name, department_id, d.location_id
FROM HR.employees e
JOIN HR.departments d USING (department_id);
```

Snapshot:



The screenshot shows a SQL query execution interface. At the top, the query is displayed in a text area. Below the query, there is a tab labeled "Query Result" with a close button. Under the tab, there are icons for a bookmark, a document, a refresh, and a SQL icon, followed by the text "Fetched 50 rows in 0.012 seconds". The results are shown in a table with three columns: FULL_NAME, DEPARTMENT_ID, and LOCATION_ID. The table contains 24 rows of data, numbered 1 through 24.

	FULL_NAME	DEPARTMENT_ID	LOCATION_ID
1	Jennifer Whalen	10	1700
2	Pat Fay	20	1800
3	Michael Hartstein	20	1800
4	Sigal Tobias	30	1700
5	Karen Colmenares	30	1700
6	Shelli Baida	30	1700
7	Den Raphaely	30	1700
8	Alexander Khoo	30	1700
9	Guy Himuro	30	1700
10	Susan Mavris	40	2400
11	Kevin Feeney	50	1500
12	Jean Fleaur	50	1500
13	Adam Fripp	50	1500
14	Timothy Gates	50	1500
15	Ki Gee	50	1500
16	Girard Geoni	50	1500
17	Douglas Grant	50	1500
18	Vance Jones	50	1500
19	Payam Kaufling	50	1500
20	Renske Ladwig	50	1500
21	Britney Everett	50	1500
22	Jennifer Dilly	50	1500
23	Julia Dellinger	50	1500
24	Curtis Davies	50	1500

Q6)

Query:

```
SELECT first_name, last_name, department_name
FROM HR.employees
NATURAL JOIN HR.departments;
```

Snapshot:

Worksheet

Query Builder

```

SELECT first_name, last_name, department_name
FROM HR.employees
NATURAL JOIN HR.departments;

```

Query Result x

SQL | All Rows Fetched: 32 in 0.004 seconds

	FIRST_NAME	LAST_NAME	DEPARTMENT_NAME
1	Neena	Kochhar	Executive
2	Lex	De Haan	Executive
3	Bruce	Ernst	IT
4	David	Austin	IT
5	Valli	Pataballa	IT
6	Diana	Lorentz	IT
7	Daniel	Faviet	Finance
8	John	Chen	Finance
9	Ismael	Sciarra	Finance
10	Jose Manuel	Urman	Finance
11	Luis	Popp	Finance
12	Alexander	Khoo	Purchasing
13	Shelli	Baida	Purchasing
14	Sigal	Tobias	Purchasing
15	Guy	Himuro	Purchasing
16	Karen	Colmenares	Purchasing
17	Laura	Bissot	Shipping
18	Mozhe	Atkinson	Shipping
19	James	Marlow	Shipping
20	TJ	Olson	Shipping
21	Peter	Tucker	Sales
22	David	Bernstein	Sales
23	Peter	Hall	Sales
24	Christopher	Olsen	Sales

Q7)

Query:

```
SELECT e.first_name AS employee_name, m.first_name AS manager_name
FROM HR.employees e
JOIN HR.employees m ON e.manager_id = m.employee_id;
```

Snapshot:

Worksheet

Query Builder

```
SELECT e.first_name AS employee_name, m.first_name AS manager_name
FROM HR.employees e
JOIN HR.employees m ON e.manager_id = m.employee_id;
```

Query Result x

SQL | Fetched 50 rows in 0.013 seconds

	EMPLOYEE_NAME	MANAGER_NAME
1	William	Gerald
2	Lisa	Gerald
3	Sundita	Gerald
4	Tayler	Gerald
5	Harrison	Gerald
6	Elizabeth	Gerald
7	Alexander	Lex
8	Clara	Alberto
9	Mattea	Alberto
10	David	Alberto
11	Danielle	Alberto
12	Amit	Alberto
13	Sundar	Alberto
14	Nandita	Adam
15	TJ	Adam
16	James	Adam
17	Julia	Adam
18	Anthony	Adam
19	Alexis	Adam
20	Laura	Adam
21	Mozhe	Adam
22	Jose Manuel	Nancy
23	Ismael	Nancy
24	Luis	Nancy

Q8)

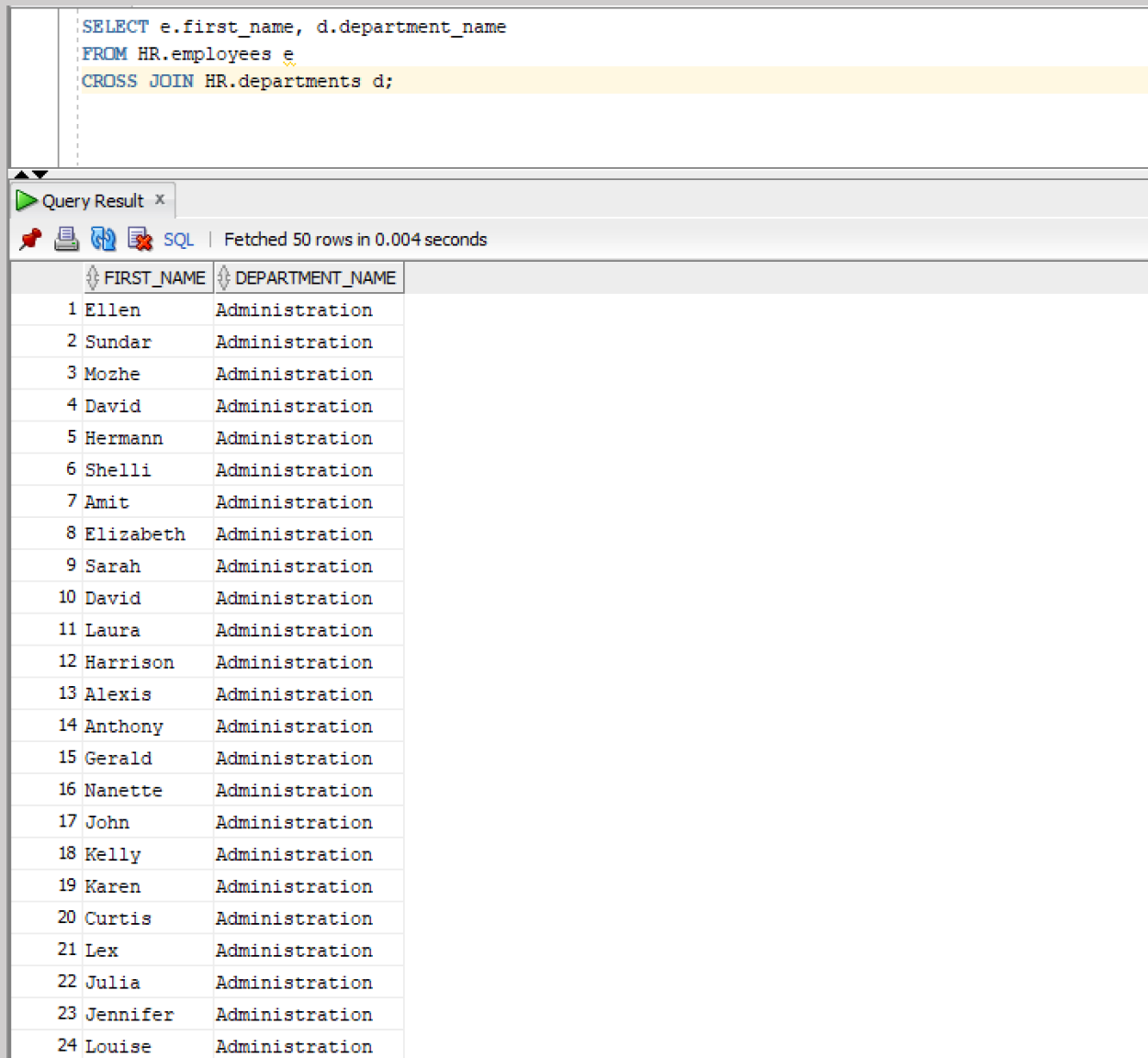
Query:

```
SELECT e.first_name, d.department_name
```

```
FROM HR.employees e
```

```
CROSS JOIN HR.departments d;
```

Snapshot:



The screenshot shows a SQL query execution interface. At the top, the query is displayed in a text area: `SELECT e.first_name, d.department_name FROM HR.employees e CROSS JOIN HR.departments d;`. Below the query, a tab labeled "Query Result" is active. The status bar indicates "Fetched 50 rows in 0.004 seconds". The results are displayed in a table with two columns: "FIRST_NAME" and "DEPARTMENT_NAME". The table contains 24 rows, all of which have the department name "Administration".

	FIRST_NAME	DEPARTMENT_NAME
1	Ellen	Administration
2	Sundar	Administration
3	Mozhe	Administration
4	David	Administration
5	Hermann	Administration
6	Shelli	Administration
7	Amit	Administration
8	Elizabeth	Administration
9	Sarah	Administration
10	David	Administration
11	Laura	Administration
12	Harrison	Administration
13	Alexis	Administration
14	Anthony	Administration
15	Gerald	Administration
16	Nanette	Administration
17	John	Administration
18	Kelly	Administration
19	Karen	Administration
20	Curtis	Administration
21	Lex	Administration
22	Julia	Administration
23	Jennifer	Administration
24	Louise	Administration

Q9)

Query:

```
SELECT e.first_name, d.department_name
```

FROM HR.employees e

FULL OUTER JOIN HR.departments d ON e.department_id = d.department_id;

Snapshot:

Worksheet

Query Builder

```

SELECT e.first_name, d.department_name
FROM HR.employees e
FULL OUTER JOIN HR.departments d ON e.department_id = d.department_id;

```

Query Result x

SQL | Fetched 50 rows in 0.008 seconds

	FIRST_NAME	DEPARTMENT_NAME
1	Steven	Executive
2	Neena	Executive
3	Lex	Executive
4	Alexander	IT
5	Bruce	IT
6	David	IT
7	Valli	IT
8	Diana	IT
9	Nancy	Finance
10	Daniel	Finance
11	John	Finance
12	Ismael	Finance
13	Jose Manuel	Finance
14	Luis	Finance
15	Den	Purchasing
16	Alexander	Purchasing
17	Shelli	Purchasing
18	Sigal	Purchasing
19	Guy	Purchasing
20	Karen	Purchasing
21	Matthew	Shipping
22	Adam	Shipping
23	Payam	Shipping
24	Shanta	Shipping

Q10)

Query:

```
SELECT e.first_name, d.department_name, j.job_title
FROM HR.employees e
JOIN HR.departments d ON e.department_id = d.department_id
JOIN HR.jobs j ON e.job_id = j.job_id;
```

Snapshot:

<pre>SELECT e.first_name, d.department_name, j.job_title FROM HR.employees e JOIN HR.departments d ON e.department_id = d.department_id JOIN HR.jobs j ON e.job_id = j.job_id;</pre>			
Query Result x			
SQL Fetched 50 rows in 0.009 seconds			
	FIRST_NAME	DEPARTMENT_NAME	JOB_TITLE
1	Jennifer	Administration	Administration Assistant
2	Pat	Marketing	Marketing Representative
3	Michael	Marketing	Marketing Manager
4	Den	Purchasing	Purchasing Manager
5	Karen	Purchasing	Purchasing Clerk
6	Guy	Purchasing	Purchasing Clerk
7	Sigal	Purchasing	Purchasing Clerk
8	Alexander	Purchasing	Purchasing Clerk
9	Shelli	Purchasing	Purchasing Clerk
10	Susan	Human Resources	Human Resources Representative
11	Matthew	Shipping	Stock Manager
12	Adam	Shipping	Stock Manager
13	Payam	Shipping	Stock Manager
14	Shanta	Shipping	Stock Manager
15	Kevin	Shipping	Stock Manager
16	John	Shipping	Stock Clerk
17	Stephen	Shipping	Stock Clerk
18	Renske	Shipping	Stock Clerk
19	Hazel	Shipping	Stock Clerk
20	Ki	Shipping	Stock Clerk
21	Michael	Shipping	Stock Clerk
22	Jason	Shipping	Stock Clerk
23	Peter	Shipping	Stock Clerk
24	James	Shipping	Stock Clerk

Q11)

Query:

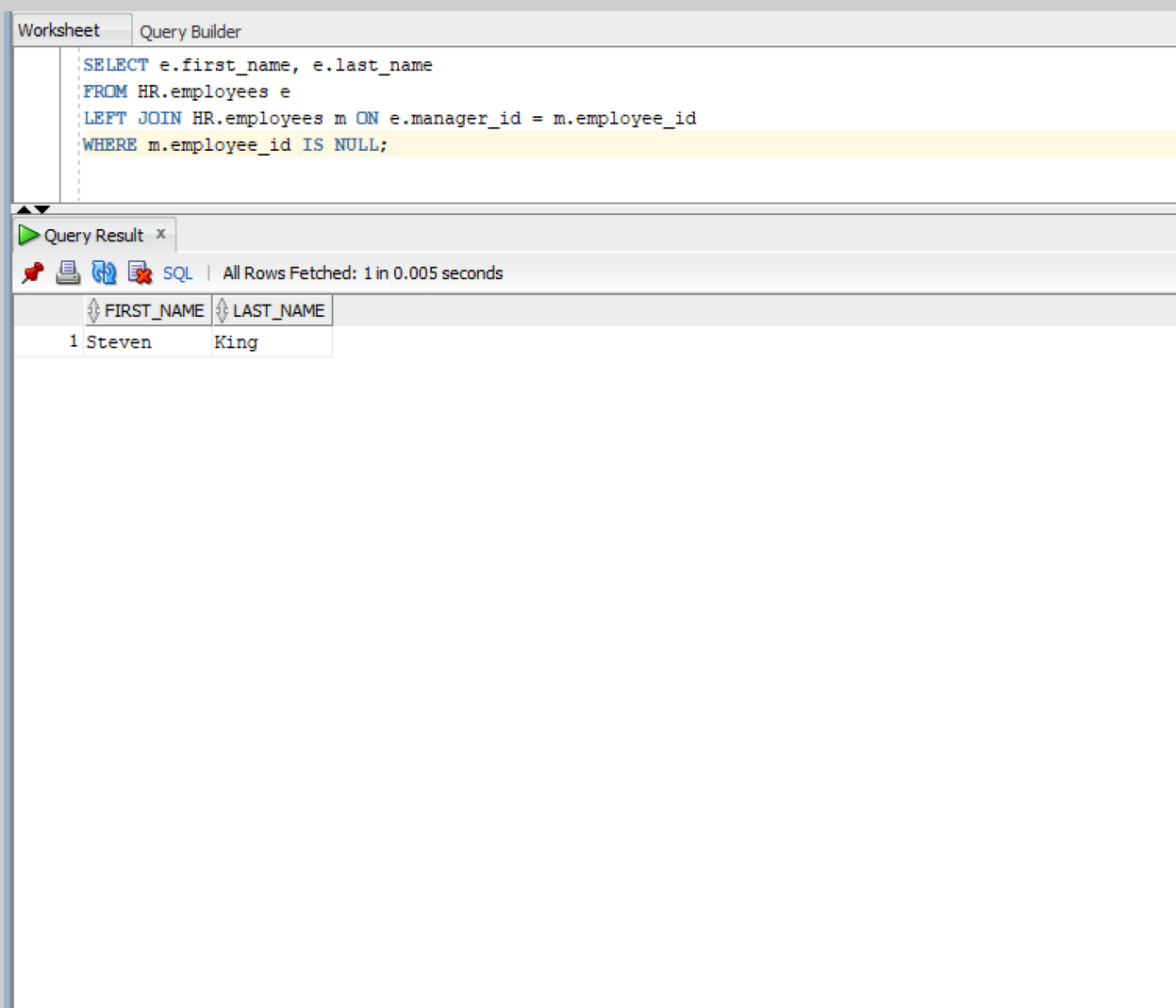
```
SELECT e.first_name, e.last_name
```

```
FROM HR.employees e
```

```
LEFT JOIN HR.employees m ON e.manager_id = m.employee_id
```

```
WHERE m.employee_id IS NULL;
```

Snapshot:



The screenshot shows a database query tool interface. At the top, there are two tabs: "Worksheet" and "Query Builder". The "Query Builder" tab is active, displaying the following SQL query:

```
SELECT e.first_name, e.last_name
FROM HR.employees e
LEFT JOIN HR.employees m ON e.manager_id = m.employee_id
WHERE m.employee_id IS NULL;
```

Below the query editor, there is a "Query Result" tab. It shows the execution status: "All Rows Fetched: 1 in 0.005 seconds". The results are displayed in a table with two columns: "FIRST_NAME" and "LAST_NAME". The table contains one row of data:

	FIRST_NAME	LAST_NAME
1	Steven	King

Q12)

Query:

```
SELECT d.department_name  
FROM HR.departments d  
LEFT JOIN HR.employees e ON d.department_id = e.department_id  
WHERE e.employee_id IS NULL;
```

Snapshot:

Worksheet Query Builder	
<pre>SELECT d.department_name FROM HR.departments d LEFT JOIN HR.employees e ON d.department_id = e.department_id WHERE e.employee_id IS NULL;</pre>	
Query Result x	
All Rows Fetched: 16 in 0.004 seconds	
DEPARTMENT_NAME	
1 Treasury	
2 Corporate Tax	
3 Control And Credit	
4 Shareholder Services	
5 Benefits	
6 Manufacturing	
7 Construction	
8 Contracting	
9 Operations	
10 IT Support	
11 NOC	
12 IT Helpdesk	
13 Government Sales	
14 Retail Sales	
15 Recruiting	
16 Payroll	

Q13)

Query:

```
SELECT employee_id FROM HR.employees
```

```
UNION
```

```
SELECT employee_id FROM HR.job_history;
```

Snapshot:

Worksheet

Query Builder

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Q14)

Query:

```
SELECT employee_id FROM HR.employees
```

```
UNION ALL
```

```
SELECT employee_id FROM HR.job_history;
```

Snapshot:

Worksheet		Query Builder	
		<pre>SELECT employee_id FROM HR.employees UNION ALL SELECT employee_id FROM HR.job_history;</pre>	
Query Result x		SQL Fetched 50 rows in 0.005 seconds	
	EMPLOYEE_ID		
1	100		
2	101		
3	102		
4	103		
5	104		
6	105		
7	106		
8	107		
9	108		
10	109		
11	110		
12	111		
13	112		
14	113		
15	114		
16	115		
17	116		
18	117		
19	118		
20	119		
21	120		
22	121		
23	122		
24	123		

Q15)

Query:

```
SELECT employee_id FROM HR.employees
```

```
UNION
```

```
SELECT employee_id FROM HR.job_history
```

```
ORDER BY employee_id;
```

Snapshot:

Worksheet

Query Builder

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Q16)

Query:

```
SELECT employee_id FROM HR.employees
```

```
INTERSECT
```

```
SELECT employee_id FROM HR.job_history;
```

Snapshot:

Worksheet

Query Builder

```
SELECT employee_id FROM HR.employees  
INTERSECT  
SELECT employee_id FROM HR.job_history;
```

▲▼

Query Result x

    | All Rows Fetched: 7 in 0.002 seconds

	EMPLOYEE_ID
1	101
2	102
3	114
4	122
5	176
6	200
7	201

Q17)

Query:

```
SELECT employee_id FROM HR.employees
```

```
MINUS
```

```
SELECT employee_id FROM HR.job_history;
```

Snapshot:

Worksheet

Query Builder

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Q18)

Query:

```
SELECT employee_id FROM HR.employees
```

```
UNION
```

```
SELECT employee_id FROM HR.job_history;
```

Snapshot:

Worksheet

Query Builder

```
SELECT employee_id FROM HR.employees
UNION
SELECT employee_id FROM HR.job_history;
```

Query Result x

   SQL | Fetched 50 rows in 0.003 seconds

	EMPLOYEE_ID
1	100
2	101
3	102
4	103
5	104
6	105
7	106
8	107
9	108
10	109
11	110
12	111
13	112
14	113
15	114
16	115
17	116
18	117
19	118
20	119
21	120
22	121
23	122
24	123